USD cash reserves: purchasing-power scenarios

User and decision

A family-office client holds a US dollar reserve for spending roughly three or six months away. When is cash needed, how does its date compare with each maturity, and how does purchasing power change by a reference Treasury bill's maturity under different inflation assumptions? Liquidity and meeting obligations take priority; beating inflation is not a universal cash requirement.

Scope and sources

Three observed series: Treasury 13-week and 26-week bank discount quotes, and BLS all-items CPI-U (not seasonally adjusted). Treasury inputs cover 2024-2026; CPI starts in 2023 to calculate subsequent year-on-year changes. Official XML/JSON snapshots are retained. Data.gov lists Treasury dataset 015-DO-021 as CC0; BLS states its data are public domain. Exact sources and reuse basis: SOURCE_REGISTER.csv.

Financial method

Convert a bank discount quote into price using actual quote-to-maturity days and a 360-day convention. Calculate nominal maturity proceeds, then divide by (1 + assumed annual inflation)^(days/365). Historical CPI is context, not a forecast. Quote-date purchase, no costs and fractional holdings are simplifications; actual execution and settlement are excluded.

Views and interaction

A timeline and inflation sensitivity chart accompany separate discount-rate and CPI history. Comparable maturity cards, a purchasing-power decomposition and a same-period zero-interest benchmark explain the result. Inputs include an optional cash date; history uses common calendar windows. Sources, dates, freshness status, formulas and limitations remain accessible.

Evaluation and handover

Eight evaluation cases cover source traces, arithmetic, interaction, missing values, freshness and reproducibility. Separate calculation checks and normal/failure maintenance trials are supplied. The original fresh-context AI audit is archived; a revised handover is pending; student independent recalculation and human accept/reject decisions remain pending. Deliverables include the local dashboard, saved data, README, skill, ontology, AI log, evaluation records and an eight-slide presentation.

Key decisions and limitations

Use official sources for traceability; align inflation and investment periods; preserve accepted files during reviewed maintenance. Exclude personal data, trading advice, taxes, FX, early-sale pricing and roll-over forecasts. A retirement-depletion simulator was rejected because its long-run assumptions would broaden the project beyond a focused cash-reserve decision.

Project status

Revision v2 review candidate, not human-accepted, prepared 27 October 2026. Development ID: dev-g00-usd-cash. Replace with instructor-assigned identifiers. The group must endorse the method and complete the documented human checks before submission. Educational project. Not investment advice.